

Figure 5, the participants have registered with the controller and received their IP addresses. In the background, the ZeroTier clients communicate with the ZeroTier root servers to find the new controller. The next step to independence is to have your own root server.

Root server

ZeroTier implements the functionality of the root server again in its ‘ZeroTier One’ application. The documentation refers to a self-hosted root server as a ‘moon’. The provider recommends using two moons that have a stable network connection and static IP addresses. The hardware requirement is minimal, so any virtual machine at a cloud provider can become a root server. Furthermore, the provider recommends that the root server does not simultaneously work as a network controller.

In Listing 2, a Debian server becomes the moon of the ZeroTier universe. Line 2 creates a world definition as well as crypto material. Line 4 lists the new root server’s IP address (here: 18.192.207.139) as the ‘stable endpoint’. If a second moon is added, it needs another entry under ‘roots’.

Listing 2: The Debian machine becomes a root server

```
cd /var/lib/zerotier-one/
zerotier-idtool initmoon identity.
public >> moon.json
sed -i -e '
s!stableEndpoints":.*!stableEndpoints": [ "18.192.207.139/9993" ]!
' moon.json
zerotier-idtool genmoon moon.json
mkdir -p moons.d/
mv 000000a0a9c4de8a.moon moons.d/
systemctl restart zerotier-one
```

The resulting moon file is signed in line 6 and stored in the subdirectory

moons.d. The file name is inspired by the world identifier, which is created in line 2. The clients will need this identifier later to find the moon. The root server is ready to use only after restarting ZeroTier One.

Finally, it is time to point the ZeroTier clients to the new universe. In the technical language of ZeroTier, the client ‘orbits’ the moon. Specify the world ID to guide the client to the new moon from Listing 2, whereby the leading zeros of the ID may be omitted:

```
root@client:~# zerotier-cli orbit
a0a9c4de8a a0a9c4de8a
200 orbit OK
```

The command ‘zerotier-cli peers’ shows if the client and moon meet. Executed on the moon, the command output should include the IP addresses of the other clients and show the role LEAF. Executed on the client, the list of planets should contain the self-hosted moon in the role MOON (Listing 3).

Listing 3: The client has made contact with the new moon

```
root@CentOS8 -> zerotier-cli peers
200 peers
<ztaddr> <ver> <role> <lat> <link>
<lastTX> <lastRX> <path>
1d71939404 1.8.0 LEAF -1 DIRECT
17771 17625 34.70.132.9/28575
61d294b9cb - PLANET 181 DIRECT
2658 2476 50.7.73.34/9993
62f865ae71 - PLANET 245 DIRECT
2658 2415 50.7.252.138/9993
778cde7190 - PLANET 137 DIRECT
7671 2510 103.195.103.66/9993
992fcf1db7 - PLANET 30 DIRECT
12680 2617 195.181.173.159/9993
a0a9c4de8a 1.8.1 MOON 30 DIRECT
2623 2623 18.192.207.139/9993
```

The ZeroTier clients and the new moon still have connections to other nodes in the ZeroTier universe. As soon as these are no

longer accessible, the remaining infrastructure continues to operate autonomously and is thus independent of external services or an Internet connection.

Technical background

ZeroTier hides its complexity behind a fancy Web interface and the ‘ZeroTier One’ software. In the background, a distributed network hypervisor operates over an encrypted peer-to-peer network. ZeroTier divides its complexity into an overlay network, which emulates an Ethernet switch for its subscribers; and an underlay network, which uses the shortest path to deliver the packets to the end devices (Figure 1).

The transport network consists of several root servers, which link the ZeroTier clients to each other and provide hints for short end-to-end connections. If the clients cannot find a suitable connection to each other, the root servers forward the packets. The root servers are operated by ZeroTier and thus make up the backbone network.

Above the transport underlay network, the Ethernet virtualisation layer serves as the overlay network. It presents the ZeroTier network as an Ethernet adapter to the operating system and the applications. For this to be successful, the overlay network must handle the typical tasks and protocols of Ethernet — ARP, NDP, broadcast, and multicast.

The network controller is responsible for the configuration and also acts as a certification authority, issuing certificates for the endpoints. The ZeroTier One software can be used as a client, network controller, or root server.

Routing

ZeroTier can not only connect individual clients, but also entire networks. The ZeroTier software